

Practical 4 - Estimating treatment effects using observational data

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

Datasets required: `kidney_example.csv` and `kidney_example_continuousY.csv`

Introduction

This practical is in two parts. Part A uses the kidney stones example data from the Notes, in which the outcome Y is binary. Part B considers data on a similar study but with a continuous outcome Y . The other variables are the same: X denotes a binary treatment and Z a binary confounder. Both data sets include 700 individuals.

Variable	Description
y	Binary: 0 = success; 1 = failure Continuous: a score on a quality of life questionnaire
x	Treatment: 0 = Surgery; 1 = Lithotripsy
z	Size of kidney stone: 0 = small; 1 = large

Aims

The aims of this session are:

- To estimate conditional treatment effects for both binary and continuous outcomes.
- To estimate marginal treatment effects for both binary and continuous outcomes using standardization.
- To interpret estimates of marginal and conditional treatment effects.

Part A: Binary outcome

The data for this section are in the file `kidney_example.csv`. You can load the dataset into R using:

```
kidney <- read.csv("kidney_example.csv")
```

The data are summarised in the tables below.

	($X = 0$)	($X = 1$)	Total
Success ($Y = 0$)	273	289	562
Failure ($Y = 1$)	77	61	138
Total	350	350	700

	Small stone size ($Z = 0$)		Large stone size ($Z = 1$)	
	Surgery ($X = 0$)	Lithotripsy ($X = 1$)	Surgery ($X = 0$)	Lithotripsy ($X = 1$)
Success ($Y = 0$)	81	234	192	55
Failure ($Y = 1$)	6	36	71	25
Total	87	270	263	80

Question 1

In this question we will use logistic regression to obtain estimates of the conditional odds ratio(s)

$$\frac{\Pr(Y = 1|do(X = 1), Z = z) / \Pr(Y = 0|do(X = 1), Z = z)}{\Pr(Y = 1|do(X = 0), Z = z) / \Pr(Y = 0|do(X = 0), Z = z)}$$

for $z = 0, 1$.

- First using a logistic regression model for Y with X and Z as explanatory variables (not including their interaction).
- Using a logistic regression model for Y with X , Z and their interaction $X \times Z$ as explanatory variables.

Question 2

Using your model from Question 1(ii) obtain estimates of:

- The conditional probabilities $\Pr(Y = 1|do(X = 1), Z = z)$ and $\Pr(Y = 1|do(X = 0), Z = z)$ for $z = 0, 1$. One approach is to use the `fitted.values` in your fitted model object.

ii) The conditional risk differences

$$\Pr(Y = 1|do(X = 1), Z = z) - \Pr(Y = 1|do(X = 0), Z = z), \quad z = 0, 1.$$

iii) The conditional risk ratios

$$\frac{\Pr(Y = 1|do(X = 1), Z = z)}{\Pr(Y = 1|do(X = 0), Z = z)}, \quad z = 0, 1.$$

Discuss: Compare your results from the logistic regression models with those given in Table 4.2 in the Notes (which were not obtained using regression). Interpret your conditional treatment effect estimates obtained in questions 1 and 2. What assumption do we make to interpret these estimates ‘causally’?

Question 3

We will next estimate marginal treatment effects.

- i) Using the results from your model in Question 1(ii), and information about $\Pr(Z = z)$, use the standardization formula in (4.8) to estimate $\Pr(Y = 1|do(X = 1))$ and $\Pr(Y = 1|do(X = 0))$. The standardization formula is:

$$\Pr(Y = 1|do(X = x)) = \sum_{z=0,1} \Pr(Y = 1|X = x, Z = z) \Pr(Z = z)$$

Note that the empirical/estimated probabilities of having small and large kidney stone size are $\widehat{\Pr}(Z = 0) = 0.51$ and $\widehat{\Pr}(Z = 1) = 0.49$ (check you agree!).

- ii) Using your results from (i) obtain estimates of the marginal risk difference, marginal risk ratio, and marginal odds ratio, as defined in equations (4.1)-(4.3) in the Notes.

You may find it useful to use the table below to keep track of your estimates.

$\Pr(Y = 1 do(X = 1))$	$\Pr(Y = 1 X = 1, Z = 0)$	
	$\Pr(Y = 1 X = 1, Z = 1)$	
	$\Pr(Y = 1 X = 1, Z = 0) \times \Pr(Z = 0)$	
	$\Pr(Y = 1 X = 1, Z = 1) \times \Pr(Z = 1)$	
	$\Pr(Y = 1 X = 1, Z = 0) \times \Pr(Z = 0) + \Pr(Y = 1 X = 1, Z = 1) \times \Pr(Z = 1)$	
$\Pr(Y = 1 do(X = 0))$	$\Pr(Y = 1 X = 0, Z = 0)$	
	$\Pr(Y = 1 X = 0, Z = 1)$	
	$\Pr(Y = 1 X = 0, Z = 0) \times \Pr(Z = 0)$	
	$\Pr(Y = 1 X = 0, Z = 1) \times \Pr(Z = 1)$	
	$\Pr(Y = 1 X = 0, Z = 0) \times \Pr(Z = 0) + \Pr(Y = 1 X = 0, Z = 1) \times \Pr(Z = 1)$	

Discuss: What are the interpretations of the causal marginal treatment effects you have estimated? Compare these with the conditional treatment effects estimated in questions 1 and 2.

Question 4

The `marginaleffects` package can also be used to estimate the risk difference more directly by standardization.

- i) The `avg_predictions` function can be used to obtain estimates of $\Pr(Y = 1|do(X = 0))$ and $\Pr(Y = 1|do(X = 1))$ using the same standardization procedure that you used above:

```
avg_predictions(y_xz_int, variables = "x")
```

- ii) Estimates of the marginal risk difference can be obtained using:

```
avg_comparisons(y_xz_int, variables = "x")
```

Run this. What do you conclude about the effect of lithotripsy compared to surgery on the chance of failure?

Question 5

Lastly we will use the empirical standardization method to estimate the marginal probabilities $\Pr(Y = 1|do(X = 0))$ and $\Pr(Y = 1|do(X = 1))$. This is the method that `avg_predictions` is using, but breaking down the steps involved should help consolidate your understanding of the method. Run the following code, making sure you understand what is happening at each step. Check that you get the same estimates as we obtained previously.

```
pred_covs <- subset(kidney, select=c(x,z))
#set treatment to 0
pred_covs$x <- 0
#manually calculate predicted probabilities for each individual
#use plogis(x)=exp(x)/(1+exp(x))
pred_covs$pred0 = with(pred_covs,
  plogis(-2.60269+0.7308873*x+1.607874*z-0.5245292*x*z))
#now set treatment to 1
pred_covs$x <- 1
pred_covs$pred1 = with(pred_covs,
  plogis(-2.60269+0.7308873*x+1.607874*z-0.5245292*x*z))
#calculate the mean predicted outcome in each scenario
```

```
mean(pred_covs$pred0)
mean(pred_covs$pred1)
#risk difference is
mean(pred_covs$pred1)-mean(pred_covs$pred0)
```

Part B: Continuous outcome

In the second part of this practical we will use a modified version of the kidney study data with a continuous outcome Y instead of a binary outcome. For example, Y could be a score on a quality of life questionnaire. The data are available in the file `kidney_example_continuousY.csv`. Load the data.

Question 6

What are the means of Y in each of the four groups defined by X and Z ? That is, what are $E(Y|X = x, Z = z)$ for $x, z = 0, 1$?

Question 7

Fit a linear regression of Y on X , Z and their interaction $X \times Z$. How do the coefficients relate to the means of Y in each of the four groups defined by X and Z ?

Question 8

Use your results from question 6 or 7 to estimate the conditional effect of X on Y given Z for $Z = 0, 1$, using a conditional mean difference:

$$\begin{aligned} &E(Y|do(X = 1), Z = 0) - E(Y|do(X = 0), Z = 0) \\ &E(Y|do(X = 1), Z = 1) - E(Y|do(X = 0), Z = 1) \end{aligned}$$

Discuss: interpret the estimates. What assumption do you need to make in order to interpret the conditional mean differences as causal treatment effect estimates?

Question 9

We now estimate the causal marginal treatment effect.

- i) By hand, use standardization to complete the table below, and hence to estimate the causal marginal treatment effect:

$$E(Y|do(X = 1)) - E(Y|do(X = 0))$$

Reminder, by standardization $E(Y|do(X = x))$ is given by

$$E(Y|X = x, Z = 0) \times \Pr(Z = 0) + E(Y|X = x, Z = 1) \times \Pr(Z = 1)$$

$E(Y do(X = 1))$	$E(Y X = 1, Z = 0)$	
	$\Pr(Z = 0)$	0.51
	$E(Y X = 1, Z = 1)$	
	$\Pr(Z = 1)$	0.49
$E(Y do(X = 0))$	$E(Y do(X = 1))$	
	$E(Y X = 0, Z = 0)$	
	$\Pr(Z = 0)$	0.51
	$E(Y X = 0, Z = 1)$	
	$\Pr(Z = 1)$	0.49
	$E(Y do(X = 0))$	
	$E(Y do(X = 1)) - E(Y do(X = 0))$	

- ii) You can also, as before, obtain the same estimate using the `marginalEffects` package:

```
#predicted mean outcome under setting x to 0 or 1
avg_predictions(ycts_xz_int,
  variables = "x",
  df="residual")
#and their difference
avg_comparisons(ycts_xz_int,
  variables = "x",
  df="residual")
```

Question 10

Modify the code given in question 5 to estimate the marginal mean difference treatment effect using empirical standardisation.

Question 11

Fit a linear regression of Y on X and Z , without an interaction term between X and Z , storing it in an object called `ycts_xz_noint`. What is the estimated conditional mean difference $E(Y|do(X = 1), Z = z) - E(Y|do(X = 0), Z = z)$ for $Z = 0, 1$? Compare this with the result from using:

```
avg_comparisons(ycts_xz_noint,  
                variables = "x")
```